

Track 19: Robotics & IoT

June 2026

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MR ROKESH TECHNOLOGY — Training & Internship Programs

Skill path: Electronics Basics → Sensors → Microcontrollers → IoT Protocols → Cloud → Robotics Integration

Related: [Track 21 — Raspberry Pi](#) | [Track 22 — Arduino](#) | [Track 23 — ROS2](#)

Track Overview

Who this is for: Engineering students, hobbyists, and professionals entering IoT and robotics fields.

Prerequisites: Basic physics/electronics helpful; programming basics (Python/C++) for 1-month+ tiers.

Tools: Arduino/Raspberry Pi, sensors (DHT, ultrasonic, IR), MQTT, Node-RED, Blynk/ThingSpeak, breadboards, multimeter.

Hardware kit required: Arduino Uno/Nano + sensor kit (institution-provided or student-purchased)

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	IoT awareness	Smart sensor dashboard
1 Month	80–100	Junior IoT builder	Home automation mini system
3 Months	250–300	Internship-ready	IoT + cloud capstone
6 Months	500–600	IoT developer	Multi-device IoT platform
9 Months	750–900	Robotics-IoT engineer	Integrated robot + IoT system

2-Week Crash Course

Days	Topics	Deliverable
D1–2	Electronics basics, breadboard, LEDs, resistors	Circuit lab notebook
D3–4	Sensors: temperature, motion, light	Sensor reading serial output
D5–6	Arduino/Python GPIO programming	3 sensor projects
D7–8	MQTT, Wi-Fi connectivity, cloud dashboard	Live sensor dashboard
D9–10	Integration capstone	Smart monitoring station

1-Month Foundation

Week	Topics	Project
W1	Arduino programming, digital/analog I/O	LED patterns + button control
W2	Sensors, LCD display, serial communication	Weather station (local)
W3	Wi-Fi modules, MQTT, cloud integration	Remote sensor dashboard
W4	Actuators (relay, servo), automation logic	Home automation controller

3-Month Job Ready

Month	Topics	Projects
M1	Arduino + Raspberry Pi GPIO, Python	Sensor suite projects
M2	MQTT, Node-RED, database logging	Data logger + alerts
M3	Mobile app/API integration, capstone	Smart agriculture / home IoT system

6-Month Professional

Month	Topics	Projects
M1–M3	As 3-month track	—
M4	Edge computing, OTA updates, security	Secure firmware update demo
M5	Robotics intro: motors, motor drivers	Line-following robot base
M6	Capstone	IoT-enabled robotic system

9-Month Advanced

Capstones: Industrial IoT monitoring platform + autonomous mobile robot | **Level:** L5

Cross-track: [Track 23 — ROS2](#), [Track 25 — PLC](#)

Assessment Rubric

Component	Weight	Criteria
Weekly Quiz	20%	Electronics, protocols, programming
Lab Completion	40%	Circuits work; data logged correctly
Project	30%	Functionality, documentation, demo
Portfolio	10%	Photos, videos, wiring diagrams

Appendix: Mentor Notes

Hardware: Institution should provide or specify kit list. Emphasize electrical safety. Document all wiring with Fritzing diagrams.